

Question			Marking details	Marks available																																					
				AO1	AO2	AO3	Total			Maths	Prac																														
5	(a)		Distance moved by mass P in 1 period, T (i.e. in one rotation) = $2\pi R$ and speed = distance / time = $\frac{2\pi R}{T}$ Alternative: $\omega = \frac{2\pi}{T}$ and $v = \omega R$		1		1	1	1																																
	(b)		<table><tr><th>R / m</th><th>Time of 10 rot /s</th><th>T / s</th><th>$v / \text{m s}^{-1}$</th><th>$v^2 / \text{m}^2 \text{ s}^{-2}$</th></tr><tr><td>0.50</td><td>4.7</td><td>0.47</td><td>6.68</td><td>44.6</td></tr><tr><td>0.60</td><td>5.2</td><td>0.52</td><td>7.25</td><td>52.6</td></tr><tr><td>0.70</td><td>5.6</td><td>0.56</td><td>7.85</td><td>61.6</td></tr><tr><td>0.80</td><td>6.0</td><td>0.60</td><td>8.38</td><td>70.2</td></tr><tr><td>0.90</td><td>6.3</td><td>0.63</td><td>8.98</td><td>80.6</td></tr></table> T column: All values correct (1) For column v all values correct (1) For column v^2 all values correct ecf (1) Consistent use of sig figs in each column and 2 or 3 sig figs (1)	R / m	Time of 10 rot /s	T / s	$v / \text{m s}^{-1}$	$v^2 / \text{m}^2 \text{ s}^{-2}$	0.50	4.7	0.47	6.68	44.6	0.60	5.2	0.52	7.25	52.6	0.70	5.6	0.56	7.85	61.6	0.80	6.0	0.60	8.38	70.2	0.90	6.3	0.63	8.98	80.6					4	4	4	4
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